

Advanced (Habanero)

Q1. If $f(x) = x^2$ and $g(x) = 2x$, what is $(f + g)(x)$?

- (A) $x^2 + 2x$
- (B) $x^2 - 2x$
- (C) $2x^2$
- (D) $x^2 + 2$
- (E) $2x^2 + 2$

Q2. What is the domain of $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x-1}$ when combined as $(f \cdot g)(x)$?

- (A) $(-\infty, -1) \cup (-1, \infty)$
- (B) $(-\infty, 0) \cup (0, 1) \cup (1, \infty)$
- (C) $(-\infty, 0) \cup (0, \infty)$
- (D) $(-\infty, 1) \cup (1, \infty)$
- (E) $(-1, 0) \cup (0, 1)$

Q3. Given $f(x) = x^2$ and $g(x) = x - 2$, what is $(f - g)(x)$?

- (A) $x^2 - x + 2$
- (B) $x^2 + x - 2$
- (C) $x^2 - x - 2$
- (D) $x^2 + 2$
- (E) $x^2 - 2$

Q4. If $f(x) = 2x$ and $g(x) = x^2$, what is $(f \cdot g)(x)$?

- (A) $2x^3$
- (B) $2x^2$
- (C) $x^2 + 2x$
- (D) $x^2 - 2x$
- (E) $2x^2 + x$

Q5. What is $(f + g)(x)$ if $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x+1}$?

- (A) $\frac{x+1+x}{x(x+1)}$
- (B) $\frac{x}{x+1}$
- (C) $\frac{2x+1}{x(x+1)}$
- (D) $\frac{2x}{x+1}$
- (E) $\frac{x+1}{x}$

Q6. Given $f(x) = x^2 - 4$ and $g(x) = x + 2$, what is $(f \div g)(x)$?

- (A) $\frac{x^2-4}{x+2}$
- (B) $x - 2$
- (C) $\frac{x^2+4}{x-2}$
- (D) $\frac{x^2-4}{x-2}$
- (E) $x + 2$

Q7. If $f(x) = x - 1$ and $g(x) = x + 1$, what is $(f + g)(x)$?

- (A) $2x - 1$
- (B) $2x$
- (C) $2x + 1$
- (D) $x + 1$
- (E) $x - 1$

Q8. Given $f(x) = x^2$ and $g(x) = 2x - 1$, what is $(f - g)(x)$?

- (A) $x^2 - 2x + 1$
- (B) $x^2 - 2x - 1$
- (C) $x^2 + 2x - 1$
- (D) $x^2 + 1$
- (E) $x^2 - 1$

Open Ended Questions (FRQ)

Q9. Simplify $(f + g)(x)$ where $f(x) = x^2 + 2x - 1$ and $g(x) = x^2 - 3x + 1$. Then find the domain of the resulting function.

Q10. If $f(x) = \frac{1}{x-1}$ and $g(x) = \frac{1}{x+1}$, find $(f \cdot g)(x)$. Simplify your answer and determine the domain of the combined function.

Chef's Tips

- Ensure students have all necessary materials.
- Remind students to show their work for FRQs.
- Encourage students to check their answers.